MIL-A-21577/4B(SHIPS)
15 August 1968
SUPERSEDING
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(See 6.2)

MILITARY SPECIFICATION

AMPLIFIER-OSCILLATOR GROUP AN/SIA-117

1. SCOPE

1.1 This specification covers the detailed requirements for an audio amplifier-oscillator group AN/SIA-117 for general announcing system, circuit 1MC, and aviators' announcing system, circuit 3MC, for use on aircraft carriers.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-62 - Capacitors, Fixed, Electrolytic (D.C. Aluminum, Dry Electrolytic, Polarized), General Specification for.
MIL-I-983 - Interior Communication Equipment, Naval Shipboard, Basic Design Requirements for.
MIL-C-3965 - Capacitors, Fixed, Nonsolid Electrolytic (Tantalum, Foil and Sintered-Slug) General Specification for.
MIL-A-21577 - Amplifiers, Audio Frequency; and Amplifier-Control Groups - Shipboard Announcing.

STANDARDS

MILITARY

MIL-STD-275 - Printed Wiring for Electronic Equipment.
MIL-STD-701 - Preferred and Guidance List of Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Amplifier-oscillator group AN/SIA-117 shall be in accordance with all applicable paragraphs of MIL-A-21577 and as specified herein. Whenever a requirement of MIL-A-21577 conflicts with a requirement of this specification, the requirement of this specification shall govern.

3.2 Equipment complement shall consist of the following:

(a) One common enclosure, bulkhead mounted, amplifier control rack, maximum dimensions not to exceed 36 inches high, 24 inches wide, and 21 inches deep containing:
2 Input preamplifier modules.
2 Alarm output preamplifier modules.
2 Power supply modules.
2 Oscillator modules (collision alarm).
2 Oscillator modules (chemical alarm).
2 Oscillator modules (general alarm).
4 Oscillator modules (Unassigned).
All necessary switches, meters, relays, wiring, terminals, and other components required to control three AM-2316/SIA amplifier assemblies; two connected for normal 1MC-3MC operation and the third connected for an emergency alert system.
All components required to connect to a remote AN/SIA-119 amplifier-control group to provide for its use as an emergency amplifier.

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3.3 Modular construction techniques with quick disconnect features shall be employed in the design of these assemblies to permit easy removal of the modules for servicing.

3.4 All modules shall be mounted in a vertical position to facilitate the unrestricted flow of air between modules for ventilation purposes.

3.5 Capacitors. - Tantalum electrolytic capacitors conforming to MIL-C-3965 shall be used in pre-amplifiers and oscillator assemblies. Plug-in type electrolytic capacitors may be used in power supply modules. These capacitors shall be in accordance with MIL-C-62.

3.6 Wiring. - Wiring employed shall have insulation type T as shown in the applicable table of MIL-I-983.

3.7 Semiconductor devices. - Semiconductor devices shall be used exclusively and shall be selected from MIL-STD-701.

3.8 Forced air cooling. - Forced air cooling will not be permitted.

3.9 Time constant. - Under the conditions specified in the overload limiter characteristic requirements of MIL-A-21577, the preamplifier output voltage shall return to the value specified in the overload limiter characteristic requirements within 0.025 second after the sudden increase in input signal voltage above reference level.

3.10 Printed wiring. - Where printed wiring is utilized, it shall be in accordance with standoff type lead terminations specified in MIL-STD-275.

3.11 Test facilities. - A common voltmeter with switching facilities shall be provided on the control rack to determine proper power supply voltage for amplifier-oscillators and preamplifiers. The meter calibration and circuit component values shall be such that the operating condition of the amplifiers or amplifier-oscillators can be determined directly from the meter without reference to tables or charts.

3.12 Interstage input circuit, input impedance, and input level will not be required.

3.13 Generated signals. - The amplifier-oscillator modules shall be capable of generating the following alarm signals:

Signal number	Signal function
1	Collision alarm
4	Chemical alarm
5	General alarm
7	Unassigned
8	Unassigned

3.13.1 Alarm signals numbers 2, 3, and 6 will not be required. Electronic timing shall be provided on circuits where timing is required.

3.13.2 Operation of an external alarm contact maker shall connect output of 1MC amplifier to all connected loudspeakers and output of emergency alert amplifier to emergency alert loudspeakers. Connection to all loudspeakers shall be maintained for subsequent voice announcements until system is manually reset to normal at the end of the emergency.

3.14 Output level. - The amplifier-oscillator shall be capable of maintaining an output level of not less than 10 volts into a 600 ohm resistive load for amplifier input. This voltage shall also be used as a test signal source as specified in MIL-A-21577.

3.15 Adjustment of output level. - A screw driver type adjustment with shaft-locking nut shall be provided in the amplifier-oscillator or in combination with the associated control circuits to permit independent adjustment of the amplifier-oscillator output level for the alarm signal into the amplifier input.

3.16 Input facilities. - Amplifier-oscillator group AN/SIA-117 shall have the following input facilities:

Microphone station number	Microphone station type	Circuits controlled
1 through 8	IC/MSB-2	1MC
9 through 15	IC/MSB-2	3MC

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3.17 Circuit attenuation. - Operation of the 1MC system from any microphone station shall energize a circuit that is capable of attenuating circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts, 60 cycle, single phase.

3.18 Local loudspeaker cutout circuit. - A cutout circuit shall be included for each microphone station automatically to disconnect local loudspeakers when the microphone is in use. The cutout circuit shall consist of a normally closed relay which shall operate to open the audio circuit supplying the local loudspeakers when the microphone press-to-talk switch is operated. These relays shall be installed in the amplifier control rack. Fifteen local cutout circuits are required.

3.19 Output facilities. - Amplifier-oscillator group AM/SIA-117 shall have the following output facilities:

Loudspeaker group no.	Loudspeaker group designation	Circuit
1	Officers	1MC
2	Topside	1MC
3	Crew	1MC
4	Engineering spaces	1MC
5	Flight deck walkways	1MC and 3MC
6	Hangar deck walkways	1MC and 3MC
7	Emergency Alert (Officer)	Emergency Alert
8	Emergency Alert (Crew)	Emergency Alert

3.20 Two loudspeakers groups specified in 3.19 shall be connected to each 500-watt amplifier assembly where two amplifiers are utilized. Circuit 1MC shall have priority over circuit 3MC.

3.21 Input and output metering and switching shall apply as specified in 3.22.

3.22 Combined metering. - A common volume indicator meter shall be provided for input and output metering. This meter shall indicate zero db at rated input and output power.

3.23 Test input level control. - Test input level control will not be required.

3.24 Microphone press-to-talk switches. - Reduction in output of circuit SE, ships entertainment system following operation of a press-to-talk switch, shall not apply to the 3MC system.

3.25 Visual alarm. - Operation of alarm signals numbers 1, 4, 5, 7 and 8 shall initiate operation of the visual alarm signal.

3.26 Circuit SE muting. - Operation of any alarm contact shall energize a circuit that is capable of muting circuit SE. This circuit shall be capable of making and breaking a resistive load of 7 amperes at 115 volts, 60 cycle, single phase.

3.27 Alarm signal priorities. - The order of priority for alarm signals shall be as follows:

(a) Collision alarm.
(b) Chemical attack alarm.
(c) General alarm.
(d) Unassigned.
(e) Unassigned.

The oscillator modules shall be provided with a device to prevent unauthorized or accidental changes in alarm signal priorities. This device shall allow intentional change in alarm priority by simple mechanical change.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling, examination, and tests shall be as specified in MIL-A-21577.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be as specified in MIL-A-21577.

6. NOTES

6.1 The notes specified in MIL-A-21577 are applicable to this specification.

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6.2 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN INRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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